

Experiment-2

Demonstration of Stages of Compilation

- **Preprocessing Stage:** Preprocessing is the **first stage** of compilation.

The preprocessor handles commands beginning with #, such as #include and #define. It expands header files, replaces macros, and removes comments from the source code.

For example, #include <stdio.h> is processed and the required contents of the header file are included.

Input: project2.c

Output: project2.i



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```
#include <stdio.h>
#define PI 3.14159
```

```
int main() {
    float radius, area;
    printf("Enter radius of the circle: ");
    scanf("%f", &radius);

    area = PI * radius * radius;
    printf("Area of circle = %.2f\n", area);
    return 0;
}
```

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- **Compilation Stage**: In this stage, the compiler converts the preprocessed C code into assembly language. It checks the program for syntax and other compilation-related errors.
The generated assembly code contains low-level instructions that are closer to the machine language understood by the processor.
Input: project2.i

Output:project2.s



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```
# 0 "project2.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "project2.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 415 "/usr/include/features.h" 3 4
# 1 "/usr/include/features-time64.h" 1 3 4
# 20 "/usr/include/features-time64.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 21 "/usr/include/features-time64.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 1 3 4
# 19 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 20 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 2 3 4
# 22 "/usr/include/features-time64.h" 2 3 4
# 416 "/usr/include/features.h" 2 3 4
# 523 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 1 3 4
# 730 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 731 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/long-double.h" 1 3 4
# 732 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 524 "/usr/include/features.h" 2 3 4
# 547 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 1 3 4
# 10 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/gnu/stubs-64.h" 1 3 4
# 11 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 2 3 4
# 548 "/usr/include/features.h" 2 3 4
# 34 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 2 3 4
# 29 "/usr/include/stdio.h" 2 3 4

# 1 "/usr/lib/gcc/x86_64-linux-gnu/15/include/stddef.h" 1 3 4
# 229 "/usr/lib/gcc/x86_64-linux-gnu/15/include/stddef.h" 3 4

# 229 "/usr/lib/gcc/x86_64-linux-gnu/15/include/stddef.h" 3 4
typedef long unsigned int size_t;
# 35 "/usr/include/stdio.h" 2 3 4

# 1 "/usr/lib/gcc/x86_64-linux-gnu/15/include/stdarg.h" 1 3 4
# 40 "/usr/lib/gcc/x86_64-linux-gnu/15/include/stdarg.h" 3 4
typedef __builtin_va_list __gnuc_va_list;
# 38 "/usr/include/stdio.h" 2 3 4

# 1 "/usr/include/x86_64-linux-gnu/bits/types.h" 1 3 4
# 27 "/usr/include/x86_64-linux-gnu/bits/types.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/typesizes.h" 1 3 4
```

- **Assembly Stage:** The **assembler** converts the assembly language code into **machine-level object code**. This code is stored in an object file. The object file contains machine instructions but is not yet a complete executable program because some functions and libraries may still need to be connected.

Input: project2.s

Output: project2.o

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```
.file "project2.c"
.text
.section .rodata
.LC0:
.string "Enter radius of the circle: "
.LC1:
.string "%f"
.LC3:
.string "Area of circle = %.2f\n"
.text
.globl main
.type main, @function
main:
.LFB0:
.cfi_startproc
    pushq %rbp
    .cfi_def_cfa_offset 16
    .cfi_offset 6, -16
    movq %rsp, %rbp
    .cfi_def_cfa_register 6
    subq $16, %rsp
    leaq .LC0(%rip), %rax
    movq %rax, %rdi
    movl $0, %eax
    call printf@PLT
    leaq -8(%rbp), %rax
    leaq .LC1(%rip), %rdx
    movq %rax, %rsi
    movq %rdx, %rdi
    movl $0, %eax
    call __isoc23_scanf@PLT
    movss -8(%rbp), %xmm0
    pxor %xmm1, %xmm1
    cvtss2sd %xmm0, %xmm1
    movsd .LC2(%rip), %xmm0
    mulsd %xmm0, %xmm1
    movss -8(%rbp), %xmm0
    cvtss2sd %xmm0, %xmm0
    mulsd %xmm1, %xmm0
    cvtsd2ss %xmm0, %xmm0
    movss %xmm0, -4(%rbp)
    pxor %xmm2, %xmm2
    cvtss2sd -4(%rbp), %xmm2
    movq %xmm2, %rax
    leaq .LC3(%rip), %rdx
    movq %rax, %xmm0
    movq %rdx, %rdi
    movl $1, %eax
    call printf@PLT
    movl $0, %eax
    leave
    .cfi_def_cfa 7, 8
    ret
    .cfi_endproc
.LFE0:
.size main, .-main
.section .rodata
.align 8
.LC2:
```

- **Linking Stage:** The linker combines the object file with the required libraries and other necessary program components. For example, functions such as `printf()` and `scanf()` are provided through standard libraries. After linking is completed, a final executable file is created.
Input: `project2.o` + required libraries
Output: `project2`



1.

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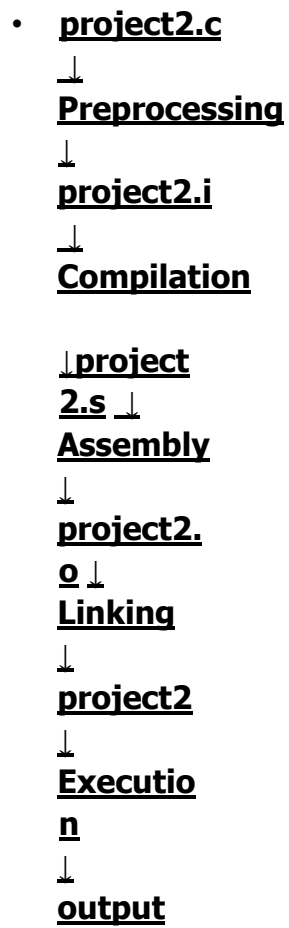
[illegible]

- **Executing Stage:** In the final stage, the operating system loads the executable program into main memory (RAM). The processor then executes the machine instructions.

For the circle-area program, the user enters the radius, and the program calculates and displays the area. Radius taken is 7

```
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(kali@kali)-[~/Desktop/avinash]
$ vi project2.c
(kali@kali)-[~/Desktop/avinash]
$ gcc -save-temps project2.c -o project2
(kali@kali)-[~/Desktop/avinash]
$ ./project2
Enter radius of the circle: 7
Area of circle = 153.94
(kali@kali)-[~/Desktop/avinash]
$ vi project2.i
(kali@kali)-[~/Desktop/avinash]
$ vi project2.s
(kali@kali)-[~/Desktop/avinash]
$ vi project2.o
(kali@kali)-[~/Desktop/avinash]
$ gcc project2.c
(kali@kali)-[~/Desktop/avinash]
$ ls -l a.out
-rwxrwxr-x 1 kali kali 16016 Aug 19 19:31 a.out
(kali@kali)-[~/Desktop/avinash]
$ ./a.out
Enter radius of the circle: 7
Area of circle = 153.94
(kali@kali)-[~/Desktop/avinash]
$ ls
a.out e2.c e2-e.i exp2.c exp2.i project2 project2.c project2.i project2.o
(kali@kali)-[~/Desktop/avinash]
$ ./a.out
Enter radius of the circle: 7
Area of circle = 153.94
(kali@kali)-[~/Desktop/avinash]
$ less project2.i
(kali@kali)-[~/Desktop/avinash]
$ less project2.s
(kali@kali)-[~/Desktop/avinash]
$ file project2.o
project2.o: ELF 64-bit LSB relocatable, x86-64, version 1 (SYSV), not stripped
(kali@kali)-[~/Desktop/avinash]
$ file a.out
a.out: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically link
(kali@kali)-[~/Desktop/avinash]
$
```



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